CS 4065/6065 Assignment 1: Investigating the Human Factors of Reaction Time

Date due: Oct 19, 2026 @ 11:59pm

Note: Revisions in document appear in **red**

Overview

In this assignment, you will explore the human factors of reaction time as it relates to choice. You will build a system to gather data, you will carry out a basic analysis, and you will draw conclusions about what factors affect the speed of reacting to a stimulus.

You will be graded on your working experimental system and report that are to be handed in through D2L.

This assignment is to be done by both **CS4065 and 6065 students and completed in groups of two**. However, CS4065 students should only work with other CS4065 students and CS6065 Students should only work with other CS6065 students.

You should also plan to participate in the experiments of other groups. In order to help everyone in the class get 6 participants for their study, please aim to participate in the data collection for 3 other assignment groups – regardless of your 4065 / 6065 status, you can participate in the data collection for any other group in the class.

Requirements

Technology

I strongly recommend using Processing for this assignment for system development (https://processing.org). However, other graphics libraries / languages are fine (e.g., p5.js, py-game)

Part 1. The system

Build a system that allows you to gather data about people's reaction time. The system should present a window (or full screen display) to the user. The system will record data about the time and errors (trying to click on a button but missing the button / clicking the incorrect choice).

Specific requirements:

- When started, the system should ask for a user number (this will be attached to all trial records). This can be done through a graphical text-entry component (e.g., a text input box) or in a settings / configuration file in your source code.
- The system should then present the testing system, the system will consist of 3, 5 or 9 buttons. One button is in the center and says "start". The other buttons will be placed equal distance from the "start" button.
- When the start button is clicked, one of the other (2, 4 or 8) buttons randomly lights up and stays lit up until the user clicks on it. The user's goal is to click the lit-up button as quickly as possible.
- Each time the user clicks the "start" button and one of the other buttons (called "target" buttons), it is considered a trial (i.e., a trial is one time that are person needs to react, you will collect data for every trials).
- For each trial, the system records the time taken from the click of the "start" button to the click of the target illuminated button. The system will also collect the number of error clicks before a successful click on the target (i.e., how many times they misclicked for the current trial attempt).
- The target buttons and are located an equal distance from the "start" button.
- Only one target button is illuminated at a time; once it is clicked it is no longer illuminated.
- The system should play a sound upon a successful click of the target button to provide feedback.
- After twenty targets, one *block* of trials is complete. The system does not present a new target, but rather shows the “click to continue” sign. This allows the user to rest before continuing.
- When the user clicks the "continue" button, the system continues to the next block of twenty trials, as described above.
- Your system will present three different blocks. Each block increases the number of target buttons.
    1. Block 1: 2 target buttons
    2. Block 2: 4 target buttons
    3. Block 3: 8 target buttons
- The system should record and write out the following data in csv format with each individual target button click (i.e., trial on a separate line):

user#, block#, trial#, elapsedTime, numberOferrors

(note that ‘elapsedTime’ is the time in milliseconds between the target being lit up and a successful click on the lit-up target)

(note that ‘numberOfErrors’ is the number of times the user clicked during this trial

without hitting the lit-up target)

(note that 'trial#" increments *within* each block; the final trial should be block 3, trial 20)

- You should log data to the console window and the data can be copied and pasted by participants into a message to you in Teams. If you prefer, you can also log the data directly to a csv file.

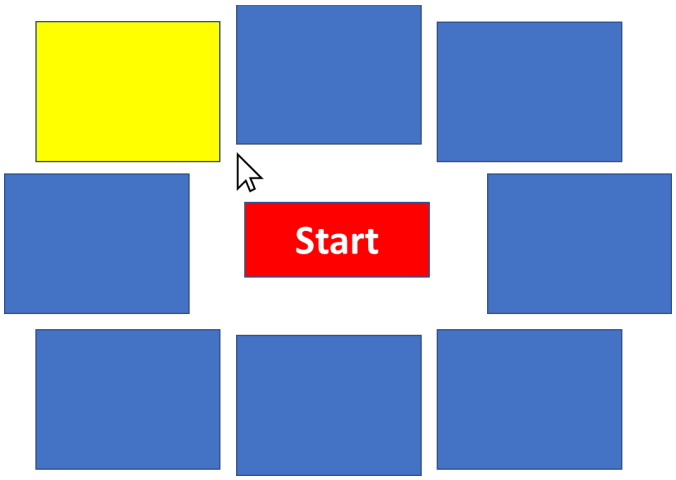


Figure 1. An example of the system after clicking start and one button is highlighted in block 3 (8 targets). Note that this was mocked up in PowerPoint.

Part 2. The investigation

Use your system to gather data from six people who are not in your group. These can be other students in the class, but can also be anyone else.

You will need to provide participants with your program and capture / log the data they produce. You will then copy and paste the data they collect into a spreadsheet like Excel and complete the steps described below.

Using a spreadsheet program (e.g., Excel), analyze the data in the following two ways:

Analysis 1 - Per-Trial Analysis

Create separate scatter plots to present the 2 dependent variables (reaction time, error count) for each of the 3 experiment conditions completed (2, 4, and 8 targets). This will result in 6 plots total (e.g., reaction time for 2 targets, error count for 2 targets, reaction time for 4 targets, …). In all cases, trial_id should be presented on the x-axis, and the dependent variable (reaction time or error count) should be presented on the y-axis. Data for all 6 participants should appear on each plot (i.e., 6 data points should appear at each trial_id / x-axis value).

This data presentation can be useful for making several types of observations. For example:

- Were there specific trails that were drastically different from others? (In the future, we will learn about how to deal with these *outlier* data points)
- Did performance evolve in a systematic way over the course of the 20 trials?
- Were there systematic differences between participants?

… you can comment on any observations in your report (see below).

Analysis 2 – Per-Condition Analysis
Create bar charts for each dependent variable (reaction time, error count; 2 charts total) to present *average* performance by experiment condition (i.e., separate bars in each chart for 2, 4, and 8 targets). Each chart should contain 3 bars, with experiment condition on the x-axis and the dependent variable on the y-axis. To accomplish this, you will have to compute a single average score for each dependent variable across the 20 trails and 6 participants. You may feel as though calculating such an average value over-simplifies the data and/or results in information loss – this is a good intuition – we will explore tools for describing and analyzing between-trial and between-participant variance later in the course.

This data presentation allows us to make other observations about the collected data:
- Were there systematic differences between experiment conditions?
- (If so,) Do these differences agree with your prediction based on Choice Reaction Time theory?

…you can comment on any observations in your report (see below).

Part 3. The report

Write a 3-4 page report (including figures) organized as detailed below:

Format of the report:

- A title and name of the report authors should be included at the top of the report.
- 1-inch margins, single spaced text
- All paragraph text in 12-point Times Roman
- Headings in 12-point bold Arial or Helvetica

1. Introduction (1/2 page max)

- State what you set out to investigate
- State why it might be important to study reaction time and how it might inform the design of a user interface based on your knowledge of typical user interfaces (i.e., how would understanding more about reaction time inform the design of

user interfaces? What type of user interfaces would this information be important for?)

2. Methods (1/2 page max, plus figures)

- Summarize the system you built, and the steps you took to gather and analyze your data
- Include a screenshot or two from your system to show how it works

3. Results (3/4 page max, plus charts)

- Display and describe your charts in your report
- Report on the what you found out – using your data, answer the question "what factor(s) (if any) affect reaction time?"
- Report on the what you found out – using your data, answer the question "what factor(s) (if any) affect the number of errors?"
- Identify and state the factors that you have discovered that affect reaction time and errors, and indicate what effect they have

4. Reflection on Results (1/2 page max)

- Reflect on the results that you found. What do you think about your results? Do they make sense? Can you provide some reasoning for why you found these results (do some careful thinking about this and try to describe why you think you found these results)?
- You can optionally do some research for this step, but it is not expected (e.g., search online databases or Wikipedia) for an answer, or ask someone who might know.

5. Implications for Design of Interfaces and Future Work (1/2 page max)

- Based on what you found, do you think that your results could be used in any user interface? How could it be used or what types of situations could it be used in?
- How confident are you in the results you found? Is there anything else that could be done to improve your results (it's OK if you think that something else could be done to improve your experiment or system)?
- Is there any information that is missing from your experiment that you think could provide further useful details for interface design? What other experiments might you do, if so?

What to hand in

- A zipped project archive with your system, including all instructions on how to run the system (if they are needed)

- A PDF version of your report
- An excel file containing all of your data and analysis

Hand in your work to D2L using the links below. Only one submission per group is need, just make sure both group members names are in the report.

Evaluation

Marks will be given for producing a system that meets the requirements above, for collecting the data required for the investigation, for carrying out a thoughtful analysis of the data with coherent charts to demonstrate your findings, and for writing a clear, thorough, and insightful report on your investigation. Additionally, your group will be penalized if you were not willing and helpful participants in the other group projects.